

Exercise 3: Agile Project Setup

Gesture Language Translator

38. Agile Board Setup

Tool: Trello (free) or Jira (free tier) | Board Name: Gesture Language Translator

Board ID	Column Name	Description	WIP Limit
COL-01	Backlog	All user stories not yet started	No limit
COL-02	To Do	Stories selected for current sprint	10
COL-03	In Progress	Stories currently being worked on	3
COL-04	In Review	Stories completed and under self-review	2
COL-05	Done	Stories fully completed and tested	No limit

39. Swimlanes Configuration

Swimlane ID	Swimlane Name	Work Type	Description
SWL-01	AI/ML Development	Feature	Tasks related to gesture detection and model training
SWL-02	Frontend/UI	Feature	Tasks related to interface design and display
SWL-03	Integration	Feature	Tasks connecting AI engine with UI modules
SWL-04	Testing	Quality	Unit, integration, and UI testing tasks
SWL-05	Documentation	Admin	SRS, design docs, and user manual tasks
SWL-06	Bug Fixes	Maintenance	Defects and issues found during testing

40. User Stories (INVEST Principle)

ID	Epic	User Story	Acceptance Criteria	Points
US-01	Camera Setup	As a user, I want to grant camera permission so the app can access my device camera	Camera permission prompt appears on launch; access is granted successfully	2
US-02	Camera Setup	As a user, I want to see a live video feed so I know the camera is active	Live video feed displayed within 2 seconds of granting permission	3
US-03	Gesture Detection	As a user, I want the system to detect my hand gestures so it can understand my sign language	Hand landmarks detected in real time with at least 85% accuracy	8
US-04	Gesture Detection	As a user, I want the system to detect my facial expressions so meaning is captured more accurately	Facial expression recognized alongside hand gestures	5
US-05	Translation	As a user, I want my gestures translated to	Correct text displayed within 2 seconds of gesture being performed	8

		text so hearing users can read what I am saying		
US-06	Translation	As a user, I want translated text to update in real time so the conversation flows naturally	Text refreshes with each new gesture without manual input	5
US-07	Text-to-Speech	As a hearing user, I want translated text read aloud so I can listen without looking at the screen	TTS plays correct audio output after each translation	5
US-08	Text-to-Speech	As a user, I want to toggle TTS on or off so I can choose my preferred mode	TTS toggle button works correctly and retains state during session	2
US-09	Session Control	As a user, I want to pause the translation so I can take a break without closing the app	Pause button stops translation; resume button restarts it	3
US-10	Session Control	As a user, I want to stop the translation session so the camera is released	Stop button ends session and releases camera access	2
US-11	History	As a user, I want to view my translation history so I can refer back to previous gestures	All translations from current session listed in order	5
US-12	History	As a user, I want to clear my translation history so my session data is removed	Clear button removes all history entries from screen	2
US-13	UI/UX	As a user, I want a simple and clean interface so I can use the app without confusion	Interface has clear labels, buttons, and readable text output	3
US-14	UI/UX	As a user, I want the app to work on both mobile and desktop so I can use any device	App renders correctly on screen sizes from 375px to 1920px wide	5
US-15	Error Handling	As a user, I want to see an error message if my gesture is not recognized so I know to try again	Unrecognized gesture shows a clear on-screen message	3
US-16	Error Handling	As a user, I want to be notified if the camera is unavailable so I understand why the app is not working	App displays camera error message if permission denied or hardware missing	2
US-17	Performance	As a user, I want the translation to complete within 2 seconds so communication feels natural	Translation latency under 2 seconds in 95% of test cases	5
US-18	Accessibility	As a deaf user, I want the interface fully operable	All critical functions work without any sound or audio dependency	3

		without audio so I am not excluded		
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41. Epic Creation

Epic ID	Epic Name	Description	Related Stories
EP-01	Camera Setup	All features related to camera access and live video display	US-01, US-02
EP-02	Gesture Detection	Hand gesture and facial expression detection using AI/ML	US-03, US-04
EP-03	Translation Engine	Converting recognized gestures into real-time text	US-05, US-06
EP-04	Text-to-Speech	Audio output of translated text	US-07, US-08
EP-05	Session Control	Pause, stop, and manage translation sessions	US-09, US-10
EP-06	Translation History	Storing and viewing past translations	US-11, US-12
EP-07	UI/UX	Interface design, responsiveness, and usability	US-13, US-14
EP-08	Error Handling	Managing unrecognized gestures and hardware errors	US-15, US-16
EP-09	Performance & Accessibility	Speed and inclusivity requirements	US-17, US-18

42. Epic Breakdown

Epic ID	Epic Name	Story ID	User Story Summary
EP-01	Camera Setup	US-01	Grant camera permission
EP-01	Camera Setup	US-02	Display live video feed
EP-02	Gesture Detection	US-03	Detect hand gestures
EP-02	Gesture Detection	US-04	Detect facial expressions
EP-03	Translation Engine	US-05	Translate gesture to text
EP-03	Translation Engine	US-06	Real-time text update
EP-04	Text-to-Speech	US-07	Read translated text aloud
EP-04	Text-to-Speech	US-08	Toggle TTS on or off
EP-05	Session Control	US-09	Pause translation
EP-05	Session Control	US-10	Stop translation session
EP-06	Translation History	US-11	View translation history
EP-06	Translation History	US-12	Clear translation history
EP-07	UI/UX	US-13	Clean and simple interface
EP-07	UI/UX	US-14	Mobile and desktop support
EP-08	Error Handling	US-15	Unrecognized gesture message
EP-08	Error Handling	US-16	Camera unavailable message
EP-09	Performance & Accessibility	US-17	Translation under 2 seconds
EP-09	Performance & Accessibility	US-18	Fully audio-free operation

43. Story Mapping Workshop

The story mapping workshop was conducted to organize all user stories by user activity and release priority. Stories were grouped by the activity the user is performing, then split into Release 1 MVP and Release 2 Enhancements.

Activity	User Goal	Release 1 - MVP	Release 2 - Enhancement
Launch App	Open the application	US-01	-
Setup Camera	Access and display camera	US-02	-
Detect Gestures	Recognize hand and face	US-03	US-04
Translate	Convert gesture to text	US-05	US-06
Output	Display or speak translation	US-07	US-08
Control Session	Manage translation flow	US-09, US-10	-
Review History	View past translations	-	US-11, US-12
Handle Errors	Notify user of issues	US-15, US-16	-
Performance	Ensure speed and accessibility	US-17	US-18

44. Story Map Structure - Release 1 (MVP)

Priority	Story ID	User Story Summary	Points
1	US-01	Grant camera permission	2
2	US-02	Display live video feed	3
3	US-03	Detect hand gestures	8
4	US-05	Translate gesture to text	8
5	US-07	Read translated text aloud	5
6	US-09	Pause translation	3
7	US-10	Stop translation session	2
8	US-13	Clean and simple interface	3
9	US-15	Unrecognized gesture message	3
10	US-16	Camera unavailable message	2
11	US-17	Translation under 2 seconds	5
Total			44

45. Definition of Ready (DoR)

Criteria ID	Criteria	Description
DoR-01	User story written	Story follows the format: As a [user], I want [goal], so that [reason]
DoR-02	Acceptance criteria defined	Clear, testable conditions for story completion are documented
DoR-03	Story estimated	Story points assigned using Planning Poker
DoR-04	Dependencies identified	Any blocking stories or external dependencies are noted
DoR-05	Story is small enough	Story can be completed within one sprint

DoR-06	Design reference available	Any relevant UI or system design is attached or linked
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46. Definition of Done (DoD)

Criteria ID	Criteria	Description
DoD-01	Code implemented	All code for the story is written and functional
DoD-02	Unit tests written	At least one unit test covers the core logic of the story
DoD-03	Acceptance criteria met	All acceptance criteria from the story are verified and passing
DoD-04	No known bugs	No critical or major defects remain open for this story
DoD-05	Code reviewed	Code has been self-reviewed against the coding standards checklist
DoD-06	Documentation updated	Relevant comments and documentation are updated
DoD-07	Board updated	Story card is moved to Done column in Trello/Jira

47. Backlog Prioritization

Priority	Story ID	User Story Summary	Epic	Points	Priority Reason
1	US-01	Grant camera permission	Camera Setup	2	Core dependency for all features
2	US-02	Display live video feed	Camera Setup	3	Required before any detection can begin
3	US-03	Detect hand gestures	Gesture Detection	8	Core AI functionality of the product
4	US-05	Translate gesture to text	Translation Engine	8	Primary product value delivered to users
5	US-13	Clean and simple interface	UI/UX	3	Required for basic usability
6	US-15	Unrecognized gesture message	Error Handling	3	Essential user feedback mechanism
7	US-16	Camera unavailable message	Error Handling	2	Essential user feedback mechanism
8	US-17	Translation under 2 seconds	Performance	5	Core non-functional requirement NFR-02
9	US-09	Pause translation	Session Control	3	Basic session management for usability
10	US-10	Stop translation session	Session Control	2	Basic session management for usability
11	US-07	Read translated text aloud	Text-to-Speech	5	Key accessibility feature for hearing users
12	US-06	Real-time text update	Translation Engine	5	Enhances natural flow of communication
13	US-14	Mobile and desktop support	UI/UX	5	Broader accessibility across devices
14	US-04	Detect facial expressions	Gesture Detection	5	Enhances translation accuracy
15	US-08	Toggle TTS on or off	Text-to-Speech	2	User preference and control

16	US-18	Fully audio-free operation	Accessibility	3	Inclusivity requirement for deaf users
17	US-11	View translation history	History	5	Nice to have enhancement feature
18	US-12	Clear translation history	History	2	Nice to have enhancement feature